

# DATA STRUCTURES

## LAB PROGRAM

5. Write a C program to find Factorial of a given number using Recursion

```
#include <stdio.h>

int factorial(int n)
{
    if (n == 0 || n == 1)
        return 1;
    else
        return n * factorial(n - 1);
}

int main()
{
    int num;

    printf("Enter a number: ");

    scanf("%d", &num);

    printf("Factorial of %d = %d", num, factorial(num));

    return 0;
}
```

### Output

```
Enter a number: 5
Factorial of 5 = 120
```

```
=== Code Execution Successful ===
```